

LAB Assignment No 2

Exercise No. 1 (Interfacing with ESP32: Temp and humidity sensor)

Code:

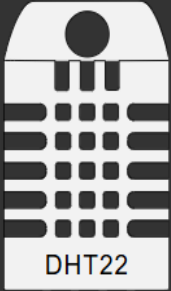
```
1  #include "DHT.h"
2  #define DHTPIN 4
3  #define DHTTYPE DHT22
4
5  DHT dht(DHTPIN, DHTTYPE);
6
7  void setup() {
8      Serial.begin(115200);
9      dht.begin();
10 }
11
12 void loop() {
13
14     float humidity = dht.readHumidity();
15     float temperature = dht.readTemperature();
16
17     Serial.print("Temperature: ");
18     Serial.print(temperature);
19     Serial.println(" °C");
20
21     Serial.print("Humidity: ");
22     Serial.print(humidity);
23     Serial.println(" %");
24
25     delay(2000);
```

Simulation Diagram:

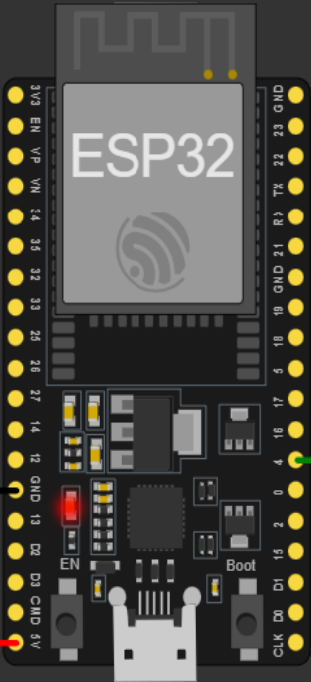
Docs

Simulation

00:51.565 59%



DHT22



ESP32



Humidity: 40.00 %

Humidity: 40.00 %
Temperature: 24.00 °C
Humidity: 40.00 %
Temperature: 24.00 °C
Humidity: 40.00 %
Temperature: 24.00 °C
Humidity: 40.00 %

Exercise No. 2 (Interfacing with ESP32: Ultrasonic Sensor)

Code:

```
1  #define trigPin 5
2  #define echoPin 18
3
4  void setup() {
5      Serial.begin(115200);
6      pinMode(trigPin, OUTPUT);
7      pinMode(echoPin, INPUT);
8  }
9
10 void loop() {
11
12     long duration;
13     float distance;
14
15     digitalWrite(trigPin, LOW);
16     delayMicroseconds(2);
17
18     digitalWrite(trigPin, HIGH);
19     delayMicroseconds(10);
20     digitalWrite(trigPin, LOW);
21
22     duration = pulseIn(echoPin, HIGH);
23
24     distance = duration * 0.034 / 2;
25
```

```
26     Serial.print("Distance: ");
27     Serial.print(distance);
28     Serial.println(" cm");
29
30     delay(1000);
31 }
```

Simulation Diagram:

Docs

Simulation

00:43.565 64%

Distance: 399.92 cm

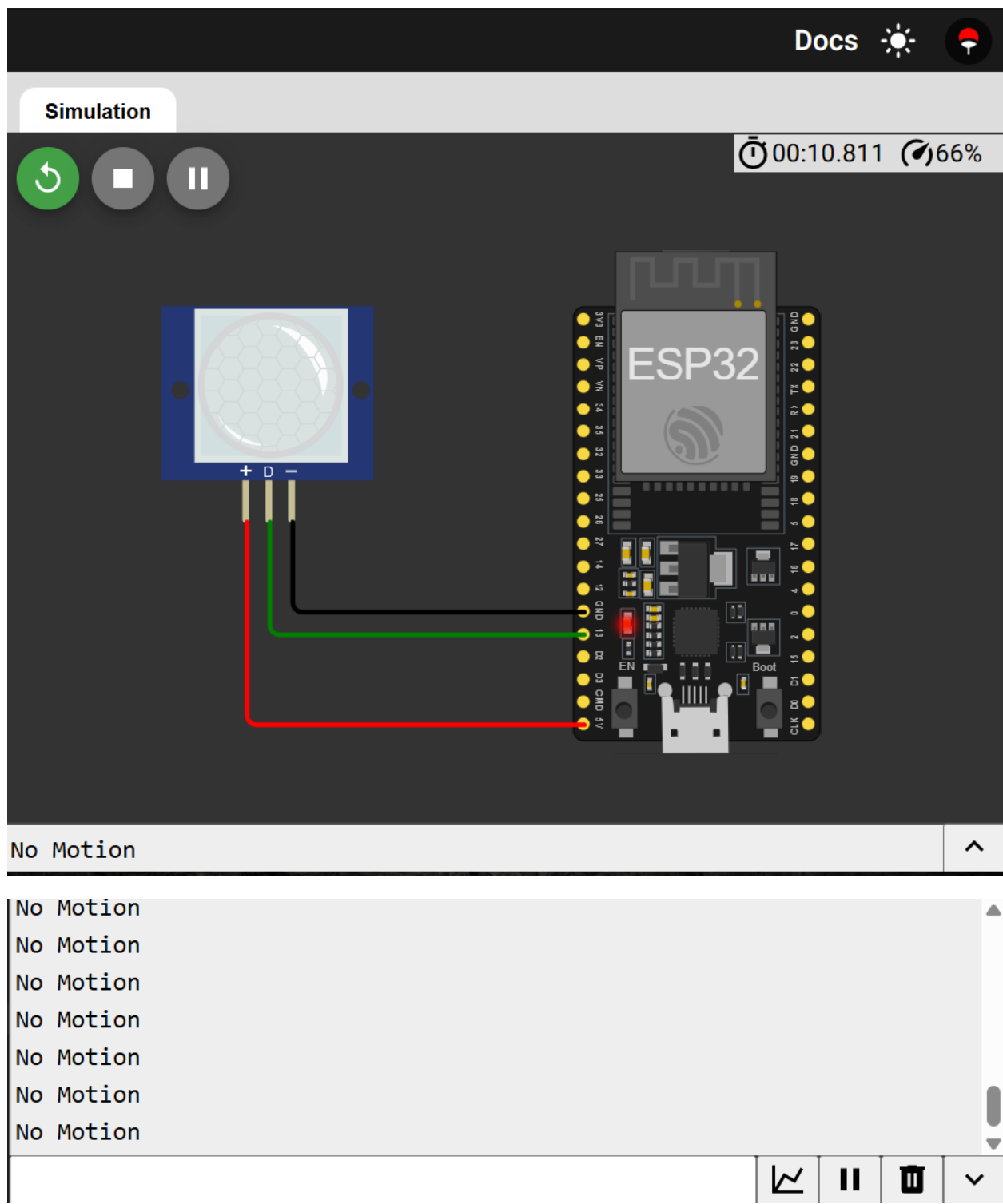
Distance: 399.92 cm
Distance: 399.92 cm
Distance: 399.92 cm
Distance: 399.92 cm
Distance: 399.92 cm
Distance: 399.92 cm
Distance: 399.92 cm

Exercise No. 3 (Interfacing with ESP32: PIR sensor)

Code:

```
1  #define PIR_PIN 13
2
3  void setup() {
4      Serial.begin(115200);
5      pinMode(PIR_PIN, INPUT);
6  }
7
8  void loop() {
9
10     int motion = digitalRead(PIR_PIN);
11
12     if(motion == HIGH)
13         Serial.println("Motion Detected");
14     else
15         Serial.println("No Motion");
16
17     delay(1000);
18 }
19
```

Simulation Diagram:



LAB Review Questions

Question 1: DHT11 with ESP32 using I2C LCD

Interface the **DHT11 Temperature and Humidity Sensor** with **ESP32** and display the temperature and humidity values on an **I2C LCD (16x2)**.

Tasks:

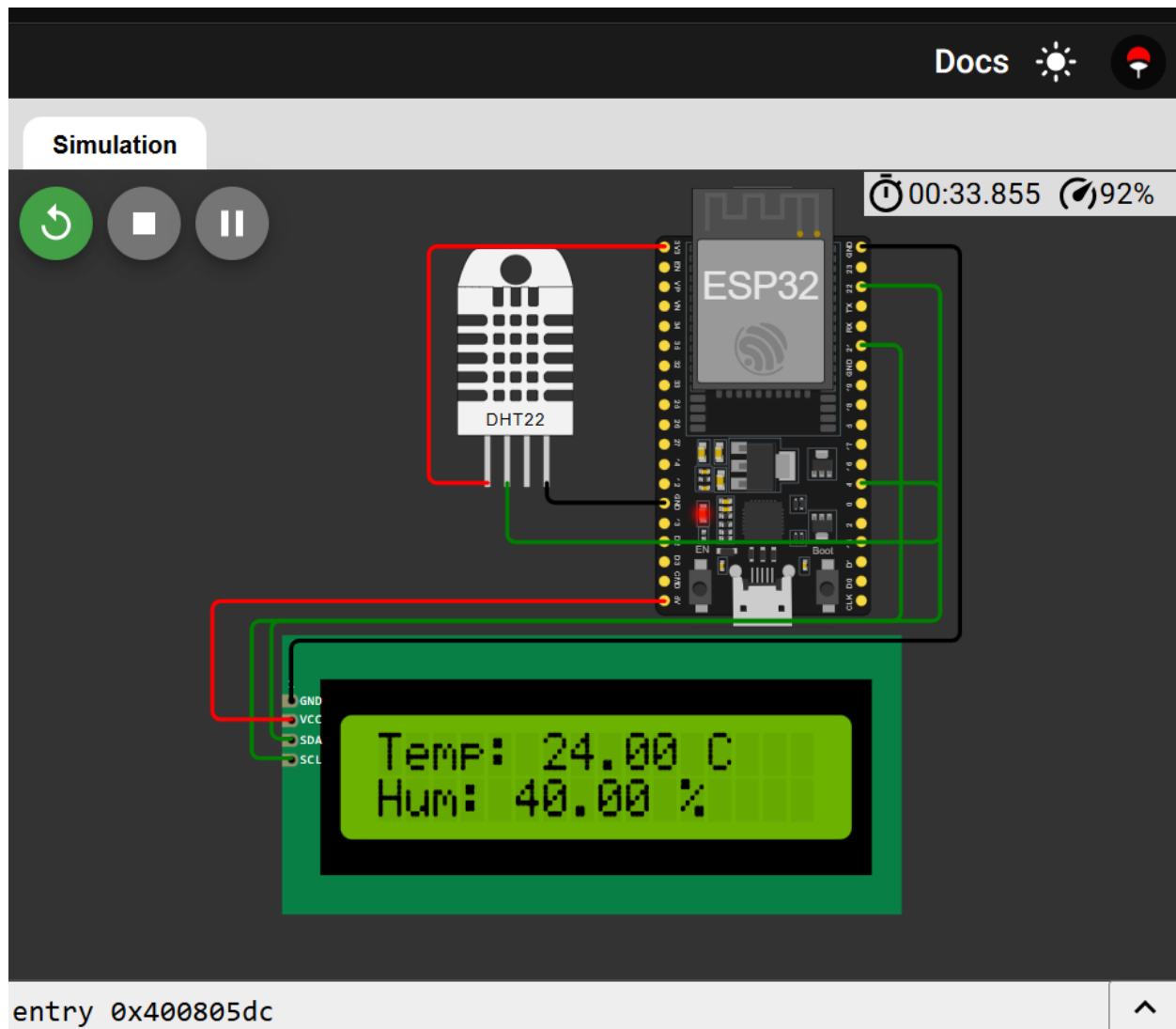
1. Write the **ESP32 program code** to read temperature and humidity from the DHT11 sensor.
2. Display the **temperature and humidity values on the I2C LCD**.
3. Draw the **complete circuit diagram / wiring diagram** of ESP32, DHT11, and I2C LCD.
4. Perform the **simulation using Wokwi or any online simulator**.
5. Capture and attach **screenshots of the running simulation and output**.

CODE:

```
1  #include <Wire.h>
2  #include <LiquidCrystal_I2C.h>
3  #include <DHT.h>
4
5  #define DHTPIN 4
6  #define DHTTYPE DHT22
7
8  DHT dht(DHTPIN, DHTTYPE);
9
10 // LCD address usually 0x27
11 LiquidCrystal_I2C lcd(0x27, 16, 2);
12
13 void setup() {
14     Serial.begin(115200);
15
16     dht.begin();
17
18     lcd.init();
19     lcd.backlight();
20
21     lcd.setCursor(0,0);
22     lcd.print("DHT11 Sensor");
23     delay(2000);
24     lcd.clear();
25 }
```

```
25     readHumidity();  
26 }  
27  
28 void loop() {  
29  
30     float humidity = dht.readHumidity();  
31     float temperature = dht.readTemperature();  
32  
33     lcd.setCursor(0,0);  
34     lcd.print("Temp: ");  
35     lcd.print(temperature);  
36     lcd.print(" C");  
37  
38     lcd.setCursor(0,1);  
39     lcd.print("Hum: ");  
40     lcd.print(humidity);  
41     lcd.print(" %");  
42  
43     delay(2000);  
44 }
```


Simulation Diagram:



Question 2: Display Data on I2C LCD using ESP32

Create a project where **ESP32** displays data on an **I2C LCD**.

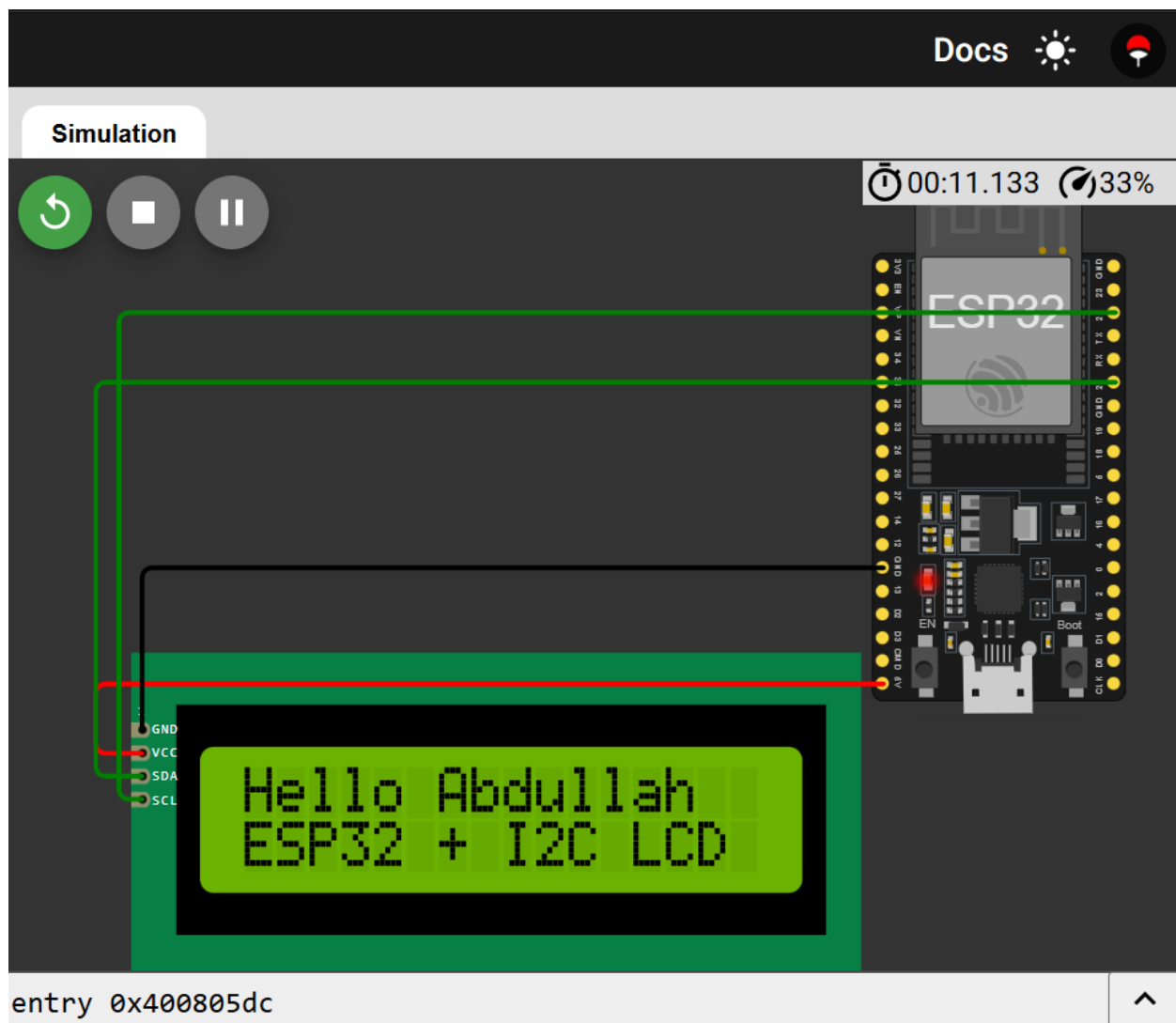
Tasks:

1. Interface **I2C LCD (16x2)** with ESP32.
2. Write a program
3. Draw the **pin connection diagram** between ESP32 and I2C LCD.
4. Run the project in **Wokwi simulation** and attach the screenshot of the output.

CODE:

```
sketch.ino • diagram.json • Library Manager libraries.txt ▼  
1  #include <Wire.h>  
2  #include <LiquidCrystal_I2C.h>  
3  
4  LiquidCrystal_I2C lcd(0x27, 16, 2);  
5  
6  void setup() {  
7  
8      lcd.init();  
9      lcd.backlight();  
10  
11     lcd.setCursor(0,0);  
12     lcd.print("Hello Abdullah");  
13  
14     lcd.setCursor(0,1);  
15     lcd.print("ESP32 + I2C LCD");  
16 }  
17  
18 void loop() {  
19  
20 }
```

Simulation Diagram:



Question 3: PIR Motion Sensor with ESP32 and I2C LCD

Interface a **PIR Motion Sensor** with **ESP32** and display the motion detection status on an **I2C LCD**.

Tasks:

1. Write ESP32 code to read the **digital output of the PIR sensor**.
2. Display the following messages on the LCD:
 - **"Motion Detected"** when motion is detected.
 - **"No Motion"** when no motion is detected.
3. Draw the **complete wiring diagram** of ESP32, PIR sensor, and I2C LCD.

4. Perform **simulation in Wokwi** and attach screenshots of the results.

CODE:

```
1  #include <Wire.h>
2  #include <LiquidCrystal_I2C.h>
3
4  #define PIR_PIN 13
5
6  LiquidCrystal_I2C lcd(0x27, 16, 2);
7
8  void setup() {
9
10     pinMode(PIR_PIN, INPUT);
11
12     lcd.init();
13     lcd.backlight();
14
15     lcd.setCursor(0,0);
16     lcd.print("PIR Sensor Ready");
17     delay(2000);
18     lcd.clear();
19 }
20
21 void loop() {
22
23     int motion = digitalRead(PIR_PIN);
24
25     if (motion == HIGH) {
```

```

27     lcd.clear();
28     lcd.setCursor(0,0);
29     lcd.print("Motion Detected");
30
31 }
32 else {
33
34     lcd.clear();
35     lcd.setCursor(0,0);
36     lcd.print("No Motion");
37
38 }
39
40 delay(1000);
41 }

```

Simulation Diagram:

